

Embedding and vector search

1. what are embeddings.
2. similarity distance
3. vector database

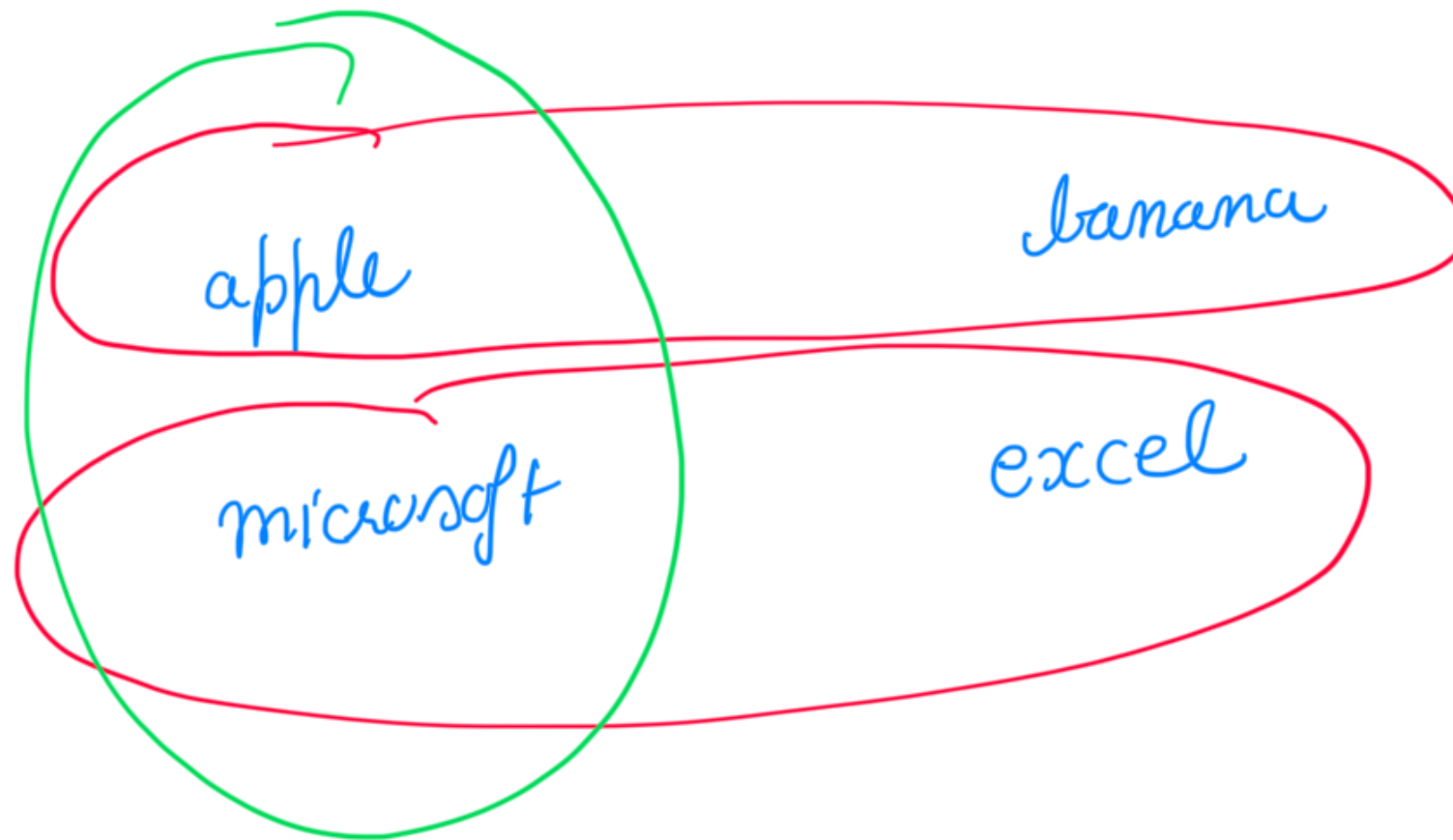
apple
↓
fruit
red

apple silicon valley
↓
company
iphone
macbook

apple is tasty

apple warranty expired

tell me apple ^{share} price



Movie recommendation system

→ similarity search

→ rating

- movie choices.
- [→ genre [action, spy, magic]
- [→ star
- [→ director

— Harry Potter

↓
magic

↗ magic
Narnia - 1

→ Spiderman - 4

← spy James Bond - 3

Lord of the Rings - 2

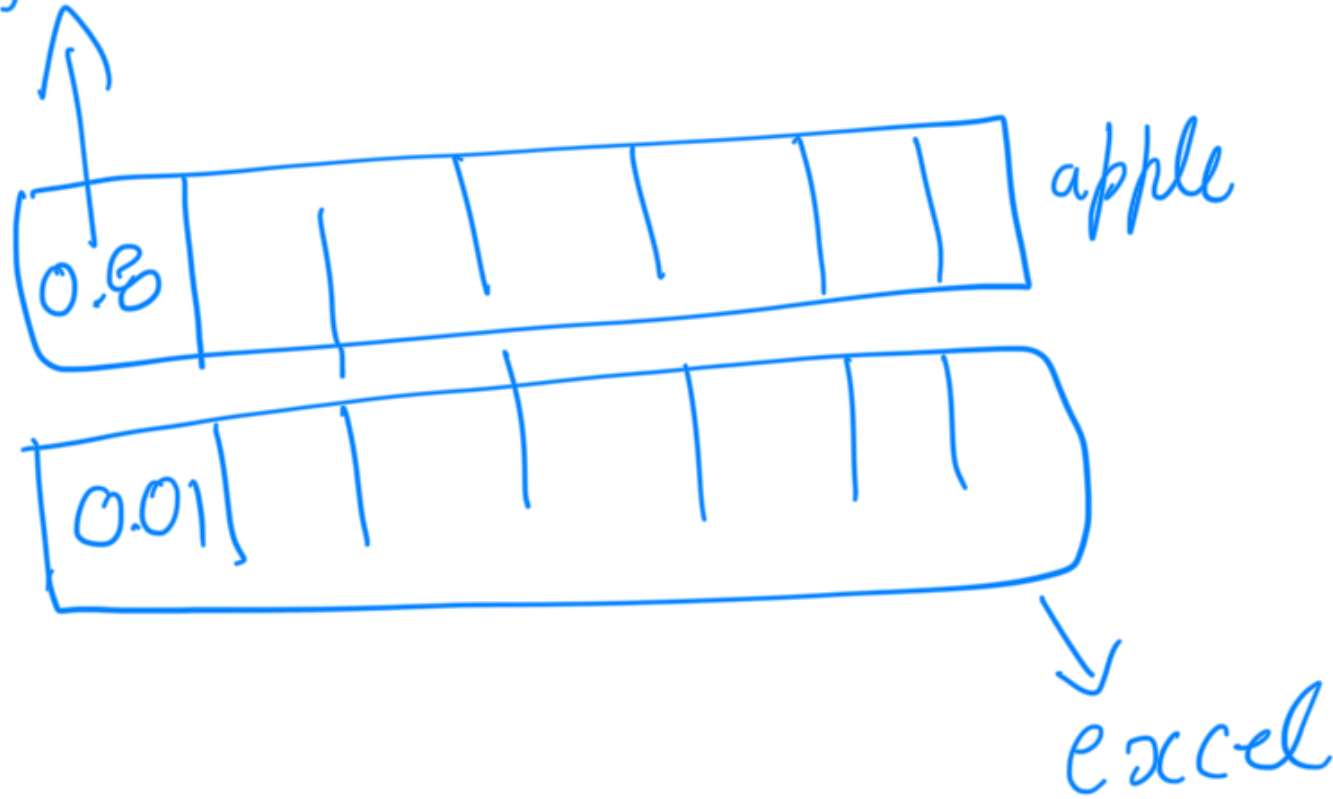
↓ magic

↑ Hidden

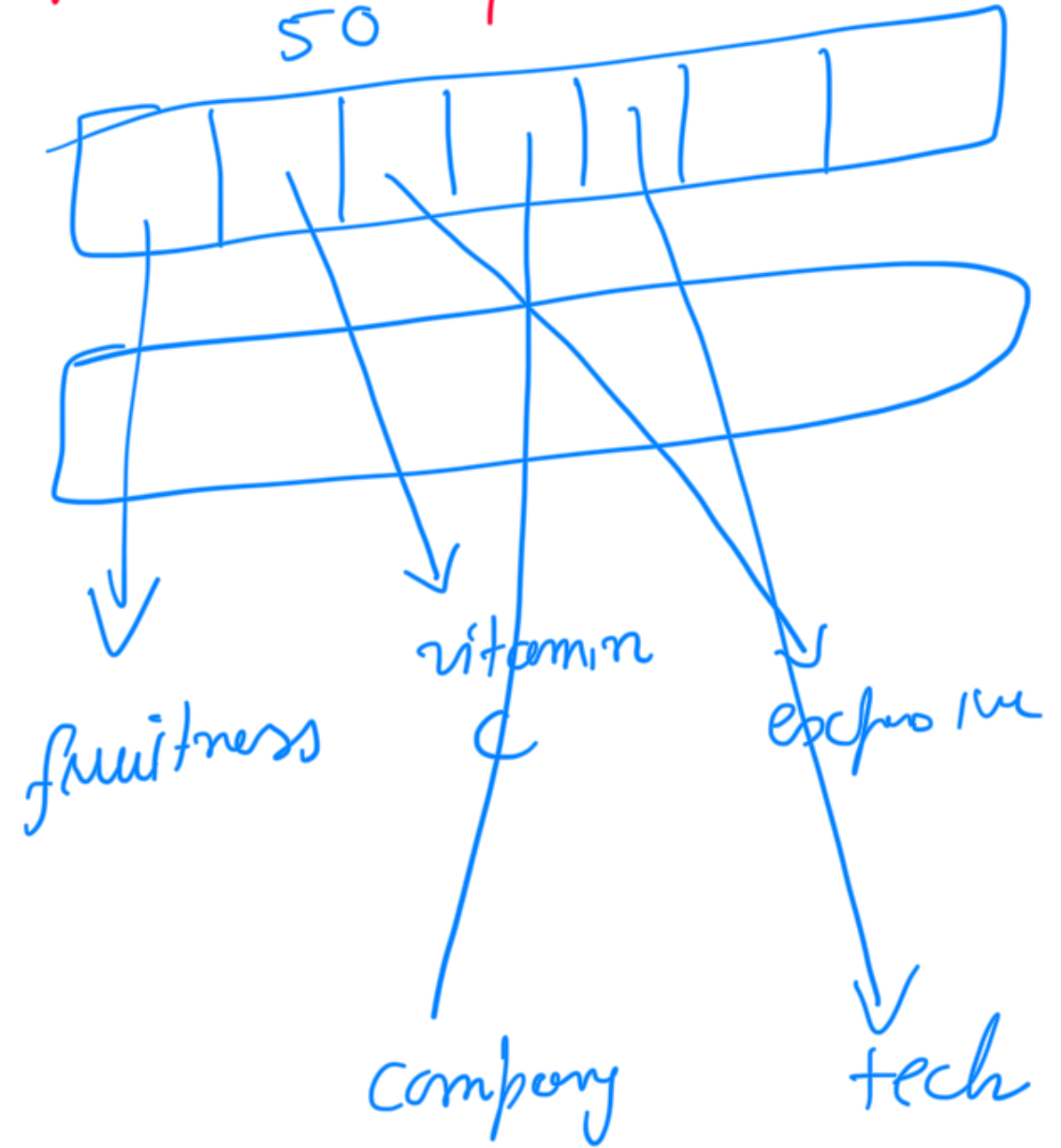
entity

embedding
model

fruitiness



numeric representation
50



keyword search



if you are less than 18 years of age, you
cannot vote

insurance -



minors cannot buy an insurance
policy.

semantic search



Q:

Can someone less than 18 years of
age have accidental coverage?

384

of
sentences

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2 similar

2 different

0

similarity
[0-1]

1
Cosine Similarity

0.7

$$\cos 0 = 1$$

$$\cos 90 = 0$$

KNN
each item

additional questions

4397.

Q: 1, 4, 5

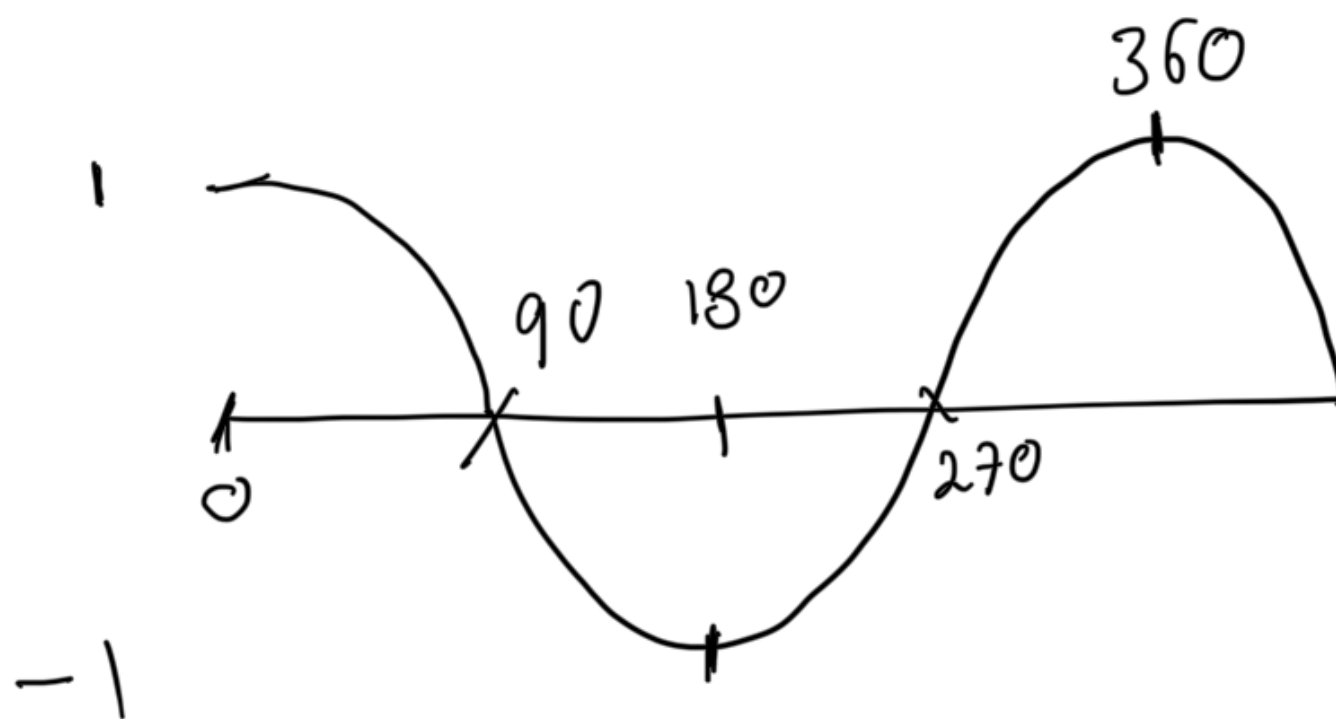
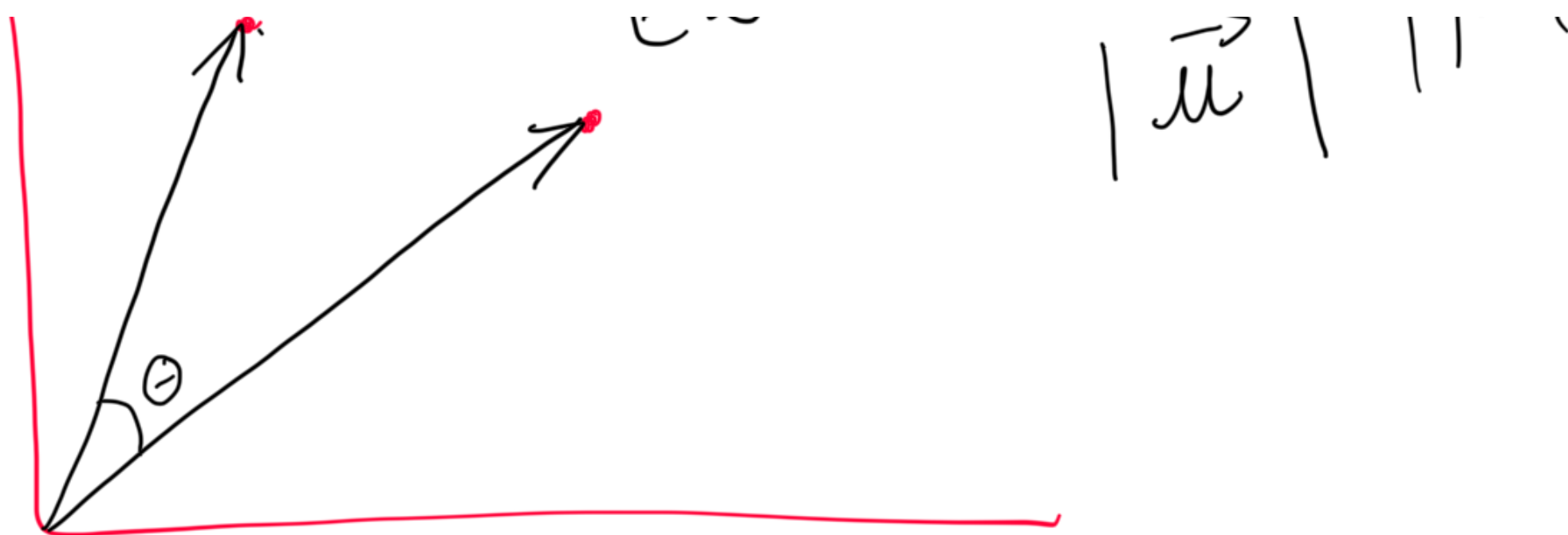
its training cats and dogs.



C2.

unit vector

$\hat{u}$



cosine
similarity

$$\vec{A} \odot \vec{B} \longrightarrow$$
$$\downarrow$$
$$[a_1, a_2, a_3, a_4] \quad [b_1, b_2, b_3, b_4]$$

$$\frac{\text{dot product} \rightarrow a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots}{|\vec{A}| |\vec{B}|}$$

$$\cos \theta = \frac{A \odot B}{|A| |B|}$$

Search

